

Sai Mun Lee

SENIOR AI/ML ENGINEER

☎ (+65) 9336 4288 | ✉ saimun92@icloud.com | 🏠 saimunlee.com | 🌐 lee-sai-mun-8a12726

Summary

Senior software engineer with 8+ years shipping production AI/ML systems end to end in enterprise SaaS, with deep production depth in multi-tenant foundation-model serving and MLOps observability. Architected and operates SAP's multi-tenant serving platform for the RPT-1 tabular foundation model on Kubernetes (10+ enterprise tenants, 8B+ recommendations served in 2025), owning inference orchestration, tenant isolation, model explainability, and synthetic-canary health monitoring for production ML pipelines. Growing hands-on depth in generative AI: built CertCoach, a hybrid-retrieval RAG system validated by a rigorous evaluation harness (deterministic IR metrics + cross-family LLM-as-judge) that caught and removed a reranker which regressed accuracy. Comfortable owning technical architecture end to end and mentoring engineers through code review and design.

Skills

LLM & Agentic AI	Agent Frameworks, Tool/Function Calling, Multi-Agent Orchestration (prototyping)
RAG & LLM APIs	RAG, Hybrid Retrieval (dense + sparse, RRF), Retrieval Evaluation (IR metrics, LLM-as-judge), Prompt Engineering, Vector Stores (pgvector), LLM APIs (Claude, OpenAI, Gemini)
ML & Serving Languages	Foundation-Model Serving, Model Inference & Optimization (latency, cost), Recommender Systems, Explainability, PyTorch
Backend & APIs	Python, Go, TypeScript, JavaScript, Bash
Backend & APIs	REST API Design, Microservices, FastAPI, Flask, Django, PostgreSQL, HANA DB, Elasticsearch
Infrastructure	Kubernetes, Docker, Cloud Foundry, Jenkins, Terraform
Cloud	AWS (Solutions Architect Pro, in progress), GCP (Associate Cloud Engineer)
Security	Multi-tenant Isolation, Encryption (at-rest & in-transit)
Practices	System Design, Distributed Systems, ML Platform Architecture, Code Review, TDD
Spoken	English, Chinese

Experience

SAP Asia

SENIOR SOFTWARE ENGINEER | CO-PRODUCT OWNER

Singapore

Dec 2020 - Present

- Architected and shipped SAP's Tabular AI platform end to end — a model-agnostic, highly scalable multi-tenant serving layer for the RPT-1 foundation model on Kubernetes (SAP AI Core), targeting 10+ enterprise tenants at 100K users each.
- Owned the full lifecycle of AI features — API contracts, model selection and inference orchestration, tenant isolation — and optimized the serving path to improve latency, throughput and cost under concurrent multi-tenant workloads.
- Built explainability tooling that surfaces feature- and cohort-level explanation signals from recommendation models, supporting model-quality reviews and stakeholder trust in model outputs.
- Designed a Postgres-native distributed-systems layer for the platform — per-tenant sharding, Postgres advisory locks as a lightweight distributed-lock server, and a Postgres-backed state-machine orchestrator sequencing per-tenant training cycles — as a deliberate alternative to standing up Redis, ZooKeeper or Airflow.
- Established CI/CD (Jenkins) with unit, integration and e2e coverage across the recommendation and inference pipelines to catch degradations before customer impact.
- Implemented enterprise security — encryption at rest and in transit, and tenant data isolation — for employee data on SAP SuccessFactors.
- Served as technical lead for Personalized Recommendations, a recommender engine generating 8B+ recommendations in 2025, and architected People Like Me, a service that aggregates customer data into training data for Personalized Recommendations and exposes its own API for consuming recommendations; set engineering best practices and mentored engineers through code review and pair programming.

TAU Express

LEAD SOFTWARE ENGINEER

Singapore

Apr. 2020 - Dec. 2020

- Led development of a knowledge-management and document-retrieval system enabling analysts to retrieve and correlate insights across cybersecurity documents via graph relationships (Neo4J) and full-text search (ElasticSearch) — retrieval foundations directly relevant to RAG.
- Owned end-to-end delivery: gathered requirements directly from users, designed the system architecture, and facilitated communication between users and the development team.
- Tech: React, Django, Docker, Neo4J, ElasticSearch, Cytoscape.js

SPIRIT Research Center (NTU)

RESEARCH ENGINEER

Singapore

Feb. 2018 - Mar. 2020

- Developed the Intelligent Case Retrieval System (ICRS) for Singapore's Smart Nation initiative — a full-stack document-retrieval application (Django, Vue.js) for efficient retrieval of legal documents.
- Built web crawlers (Selenium, BeautifulSoup) to index Chinese news articles and social media content for document-retrieval pipelines.

Projects

Personal Project

CERTCOACH — RAG CERTIFICATION-STUDY ASSISTANT

Aug 2026

- Designed a hybrid-retrieval RAG pipeline — dense vector search (pgvector) fused with Postgres full-text (sparse) search via Reciprocal Rank Fusion, plus chunk-neighbor context expansion.
- Built an evaluation harness combining deterministic IR metrics (recall@k, hit@k, MRR@k, NDCG@k) with cross-family LLM-as-judge scoring, run as a structured A/B sweep across retrieval and prompting configurations.
- Used the eval harness to catch a reranker that regressed accuracy (0.761 to 0.664) while adding roughly 200x latency, and cut it from the pipeline — a data-driven call over an assumed best practice.

Certificates

Dec 2021 **AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)

Feb 2022 **GCP Associate Cloud Engineer**, Google Cloud Platform (GCP)

May 2023 **Hashicorp Certified: Terraform Associate (003)**, Hashicorp

AWS Certified Solutions Architect - Professional, Amazon Web Services (AWS)

In Progress

Education

Nanyang Technological University (NTU)

Singapore

B.S. IN COMPUTER SCIENCE (HONOURS) — MACHINE LEARNING & DATA SCIENCE SPECIALISATION

2013 - 2017